

CHAPTER 5

GREEN SKILLS - II

INFORMATION TECHNOLOGY (402)
ARTIFICIAL INTELLIGENCE (417)

L - 1

Meaning and Definition of Sustainable Development

Sustainable development refers to meeting present needs without compromising future generations' ability to meet theirs. It aims to balance economic growth, environmental care, and social well-being. Key principles include the careful use and conservation of resources and minimizing depletion of natural resources.

The concept emerged from sustainable forest management ideas in Europe during the 17th and 18th centuries and gained prominence in the 20th century due to environmental concerns. The 1987 Brundtland Report by the United Nations popularized the term, defining it as development that meets current needs without jeopardizing future generations' capacity to meet theirs.

Importance of Sustainable Development

Sustainable development is crucial for environmental maintenance. Its significance includes:

- Proper Use of Resources: It encourages efficient resource utilization and conservation.
- Development of Positive Attitude: It changes knowledge and attitudes, fostering awareness of the responsibility to preserve natural resources as shared assets, promoting conservation efforts in both natural and social environments.
- Development Based on People's Participation: People's participation is essential for achieving sustainable development, fostering local interest in development and environmental conservation.

Importance of Sustainable Development

- Limitation of Development: Effective use of limited resources can satisfy basic needs, but overuse of non-renewable resources is depleting them globally. Development should align with carrying capacity.
- Long Lasting Development: Sustainable development focuses on economic and social progress while preserving resources for future generations. Recognizing the importance of sustainability is crucial for our survival, necessitating a balance between economic growth and ecosystem preservation.

QUESTIONS

[QUESTION] Which of the following is not the principle of sustainable development?

- (a) To carefully utilise all resources
- (b) To conserve resources
- (c) To minimise the depletion of natural resources
- (d) None of the above

[QUESTION] Sustainable development is necessary for the the environment.

[QUESTION] The idea of sustainable development gained wide acceptance due to environmental concerns in the (1980 - 1990)20th century

[QUESTION] Define sustainability.

Problems Related to Sustainable Development

- Poor management of natural resources, coupled with increasing economic activities, poses significant environmental challenges.
- Key issues revolve around renewable resources like air and water, which can only assimilate finite levels of emissions and waste. Exceeding these levels can lead to rapid ecosystem deterioration.
- Assessing the regenerative capacity of natural resources is challenging, and there is considerable uncertainty regarding environmental degradation outcomes from issues like soil erosion and atmospheric pollution.
- The impact of economic activities on the environment is continually evolving.

Problems Related to Sustainable Development

- Although rising incomes increase demand for better environmental quality and investment resources, problems can worsen despite increased income.
- Population growth will exacerbate environmental degradation in the future.
- Additionally, energy demand is expected to rise significantly, particularly in developing countries, with manufacturing outputs projected to increase sixfold by 2030.
- Meeting the food demands of a growing population will require enhancing yields on existing agricultural land while protecting fragile soils and natural habitats.

Solutions for Sustainable Development

Achieving sustainable development is possible with concentrated efforts, integrating economic, environmental, and social components. Key principles include:

- Respect for all life forms.
- Conservation of natural resources.
- Preservation of Earth's vitality and diversity.
- Improvement of human life quality.
- Change in attitudes towards the environment.

Short-Term Solutions

- Halt illegal deforestation and smuggling of forest resources.
- Balance deforestation with afforestation.
- Plan and build industrial zones to manage waste effectively.
- Implement proper waste treatment, recycling, and disposal systems.

Techniques

- Adopt rainwater harvesting, drip/sprinkler irrigation, and alternative energy sources.
- Use fewer chemical fertilizers, opting for environmentally friendly pesticides.
- Relocate polluting industries away from populated areas.

Long-Term Solutions

- Enforce policies against illegal activities.
- Launch awareness campaigns for farmers and industrialists.
- Protect ecology with taxes and fines.
- Promote sustainable agriculture practices like permaculture and crop rotation.

United Nations Sustainable Development Goals (Agenda 2030)

- 1. End poverty everywhere.**
- 2. Achieve food security and promote sustainable agriculture.**
- 3. Ensure healthy lives for all.**
- 4. Provide inclusive, quality education for all.**
- 5. Achieve gender equality and empower women and girls.**
- 6. Ensure access to water and sanitation.**
- 7. Guarantee access to sustainable energy.**
- 8. Promote inclusive economic growth and decent work.**
- 9. Build resilient infrastructure and foster innovation.**
- 10. Reduce inequality within and among countries.**

United Nations Sustainable Development Goals (Agenda 2030)

- 11. Make cities safe and inclusive.**
- 12. Ensure sustainable consumption and production.**
- 13. Combat climate change and its impacts.**
- 14. Sustainably use marine resources.**
- 15. Manage forests and halt biodiversity loss.**
- 16. Promote peaceful societies.**
- 17. Revitalize global partnerships for sustainable development.**

QUESTIONS

[QUESTION] The most significant environmental problems are associated with resources like

- (a) air
- (b) water
- (c) soil
- (d) Both (a) and (b)

[QUESTION] Awareness campaigns should be launched for

- (a) farmers
- (b) industrialists
- (c) common people
- (d) Both (a) and (b)

QUESTIONS

[QUESTION] United Nations Sustainable Development Summit goals are termed as

.....

[QUESTION] The achievement of sustainable development requires the integration of economic, and components at all levels.

[QUESTION] Write any one short-term solution related to sustainable development.

THANK - YOU

